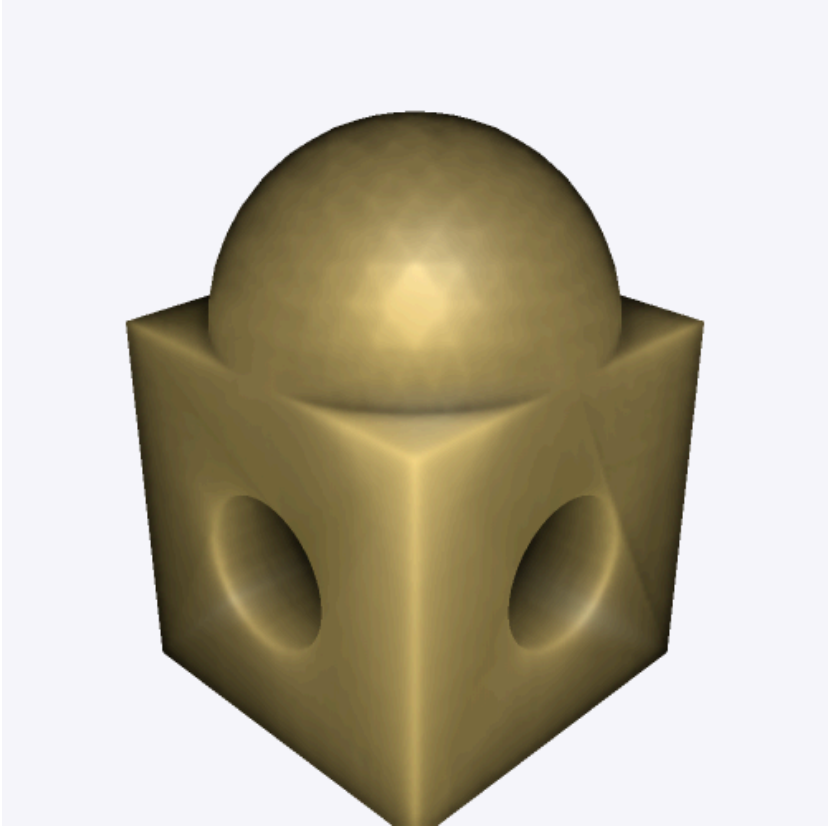
maquette-scad

Parametric CAD in Typst, via OpenSCAD + Manifold

github.com/bernsteining/maquette · bernsteining.github.io/maquette



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Introduction

maquette-scad is a Typst plugin that turns OpenSCAD-flavored geometry into meshes for `maquette` to render. Two entry points, both returning PLY bytes you feed to `render-ply`:

- **`scadypst(tree)`** — build geometry with Typst expressions (`cube`, `sphere`, `difference`, `translate`, …). The tree is a plain dict of dicts you compose with Typst's own `for` / `range` / `calc`.
- **`compile-scad(source, files?, bin?, font?, fn?, trace?)`** — hand in the text of an existing `.scad` file. Full OpenSCAD language surface: `function`, `module`, `for`, list comprehensions, `include` / `use`, `$fn` / `$fa` / `$fs`.

The CSG kernel is Manifold — every boolean is guaranteed watertight, no BSP-style open faces on non-trivial cuts.

This doc covers the maquette-scad **API**: the Typst-side entry points, their options, and the patterns for using them. It does **not** teach OpenSCAD — for that see the OpenSCAD Users Manual (language) or `crates/maquette-scad/maquette-scad/maquette-scad.typ` (per-function docstrings for every DSL helper). Once you have PLY bytes, downstream rendering (camera, lighting, materials, shadows, tone mapping) is maquette's job — see maquette's documentation. See `crates/maquette-scad/FIDELITY.md` for the exhaustive OpenSCAD language coverage tracker.

Where to find sample `.scad` files

The DSL examples below run inline — no external files needed. To exercise the `.scad` ingestion path (`compile-scad(read(...))`), grab source from:

- openscad/openscad `examples/` — the official example set that ships with the OpenSCAD editor.
- BOSL2 — comprehensive OpenSCAD utility library with hundreds of documented example fragments.
- Thingiverse (openscad tag) — large community archive; many things ship the `.scad` source alongside the `.stl`.

Quickstart

Two entry points, one plugin. Route the returned PLY to `render-ply` with whatever config you'd give any other mesh.

```
#import "@preview/maquette-scad:0.1.0": scadypst, cube, sphere, difference
#import "@preview/maquette:0.1.3": render-ply

#let part = scadypst(
  difference(
    cube(20, center: true),
    sphere(12, fn: 48),
  )
)
#render-ply(part, camera: (40, 40, 40), up: (0, 0, 1))
```

`compile-scad` is the same shape but takes source text:

```
#import "@preview/maquette-scad:0.1.0": compile-scad
#import "@preview/maquette:0.1.3": render-ply

#let part = compile-scad(read("model.scad"))
#render-ply(part)
```

Below, `show-part` is a thin wrapper this doc uses so examples can focus on the geometry without repeating render config. In real use you'd inline the `render-ply(bytes, ..)` call.

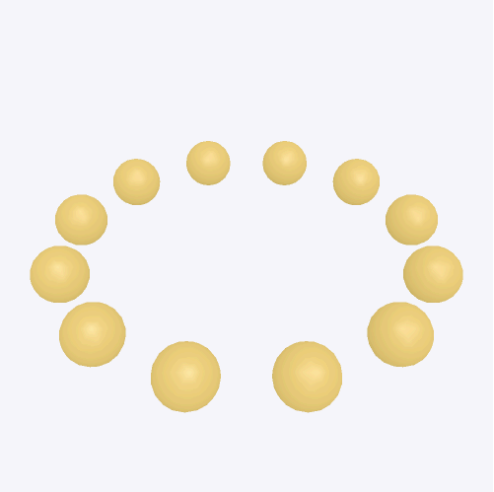
Building geometry from Typst — `scadypst`

`scadypst(tree)` compiles a tree of Typst dicts into PLY bytes. The dicts come from helpers imported from the plugin (`cube` / `sphere` / `translate` / `difference` / `hull` / `linear-extrude` / …); each helper **builds** a node in the tree, nothing runs until `scadypst` sees the whole thing.

The API mirrors OpenSCAD's own — cube, sphere, cylinder, polygon, extrusions, booleans, transforms — with Typst-native call syntax (`cube(20, center: true)` instead of `cube(20, center=true);`). For the full list see the per-function docstrings in `crates/maquette-scad/maquette-scad/maquette-scad.typ`. This doc doesn't re-document them — if you know OpenSCAD you already know the names.

The reason to reach for `scadypst` over `compile-scad(read("part.scad"))` is Typst integration: because the tree is Typst code, you get Typst's own procedural facilities for free. Here a `..for` spread inside `union` places twelve spheres on a ring:

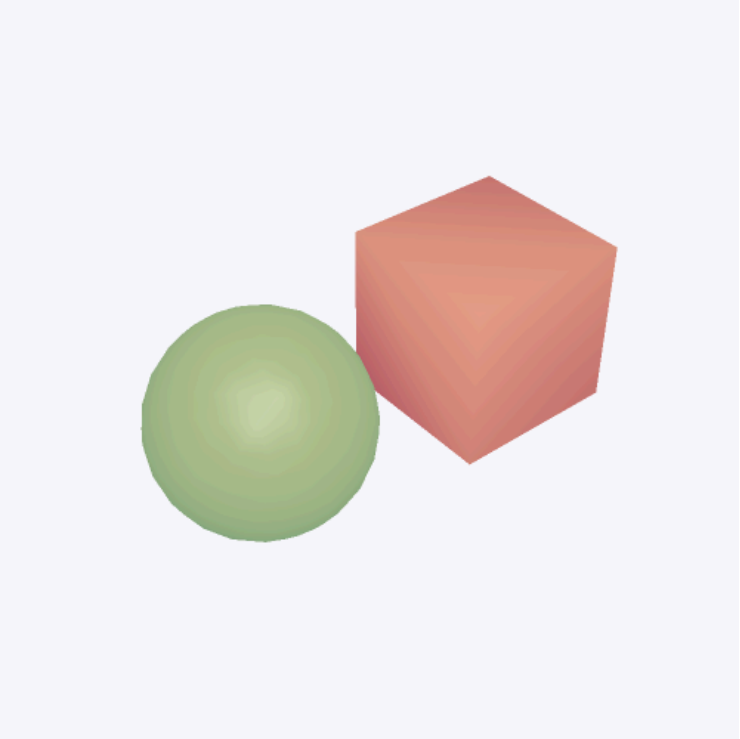
```
#let ring = union(..for i in range(12) {
  let a = i * 30
  let r = 20
  (translate(
    (calc.cos(a * 1deg) * r, calc.sin(a * 1deg) * r, 0),
    sphere(3, fn: 24),
  ),)
})
#show-part(scadypst(ring))
```



The equivalent OpenSCAD `for (i = [0:11]) { ... }` works too via `compile-scad` — but a new-in-Typst design will usually reach for Typst's iteration first, since it composes cleanly with variadic ops (`union` / `difference` / `hull` / `intersection`) via the `..` spread syntax and reads with the rest of your document code.

Colours propagate through boolean ops; the emitted PLY carries per-face RGB and `render-ply` picks it up automatically:

```
#show-part(
  scadypst(union(
    color((0.9, 0.3, 0.3), cube(10, center: true)),
    color((0.3, 0.7, 0.4), translate((15, 0, 0), sphere(6, fn:
32))),
  )),
  color: none,
)
```



Compiling `.scad` sources — `compile-scad`

For standalone `.scad` files, `compile-scad(source_text)` is all you need:

```
#let part = compile-scad(read("part.scad"))
#render-ply(part)
```

Real-world `.scad` files rarely stand alone — they `use` / `include` a library, `import` an STL/OBJ for a fastener, or ship a font for embossed text. The wasm sandbox has no filesystem, so sidecars are passed explicitly as bytes:

```
#compile-scad(read("main.scad"),
  files: (
    "utils.scad":                    read("utils.scad"),
    "gears.scad":                    read("gears.scad"),
    "MCAD/nuts_and_bolts.scad":      read("MCAD/nuts_and_bolts.scad"),
  ),
  bin: (
    "mount.stl":                     read("mount.stl", encoding: none),
  ),
  font: read("logo.ttf", encoding: none),
  fn: 48,
  trace: "trace.log",
)
```

Options:

- **`source`** (positional string) — the `.scad` text. `read("path.scad")` is the usual way to get it.
- **`files`** (dict, optional) — every `use <name>` / `include <name>` resolves against this dict. Keys are the exact names the `.scad` uses; nested paths (`MCAD/foo.scad`) are dict keys with slashes.
- **`bin`** (dict, optional) — sidecar mesh files referenced by `import()` in the source. Keys match the `import` argument. Decoded formats: STL and OBJ (DXF, 3MF and AMF are not).
- **`font`** (bytes, optional) — a TTF/OTF file for `text()` glyph rendering. One font per compile.
- **`fn`** (int, optional) — default `$fn` for the whole compile. Per-primitive `fn:` overrides.
- **`trace`** (string, optional) — dump the evaluator trace to this path. Useful when a nested `for` or `module` isn't producing what you expected.

Real-world showcase — Cyclone-PCB-Factory

For `.scad` projects big enough that the wasm plugin's 1 GB memory budget gets in the way, precompile natively and hand `render-ply` a `.ply` blob directly. `examples/scad/cyclone.typ` in the repo shows the pattern.

The Cyclone PCB Factory CNC mill — a full assembly with MCAD, obiscad, and standard_parts under `include <Cyclone.scad>` — compiles to an 11 MB mesh. Too big for the wasm plugin, so it's compiled once by the native harness at `crates/maquette-scad/examples/repro.rs` into a `.ply`, then rendered by maquette:

```
#import "@preview/maquette:0.1.3": render-ply

#let full = read("cyclone-full.ply", encoding: none)
#let openscad-view = (camera: (400, 400, 200), up: (0, 0, 1), fov: 40)
#let shot = (az, el) => render-ply(full,
  openscad-view + (azimuth: az, elevation: el, antialias: 2, zoom: 1.4),
  width: 100%,
)
#shot(35, 20)
#shot(125, 28)
```

The Manifold kernel keeps the whole assembly watertight (0 open edges) — every part renders solid from every angle even where several dozen booleans stack across the MCAD library. To reproduce: obtain `Source_files/` from the upstream repo, drop it into `examples/scad/cyclone-src/`, then

```
cargo run --release --example repro -p maquette-scad
```

which produces `cyclone-full.ply` alongside the `.typ`.

Language coverage

The `.scad` evaluator is complete for everything a typical file uses. TL;DR:

Supported (✅) — comments, numbers, bool, string, `undef`, vectors, ranges, variable assignment (last-wins scope), `let()` expression, member access, indexing, all standard operators, `$fn` / `$t` / `$preview` / `$children`, all 2D + 3D primitives, all transforms (`translate`, `rotate`, `scale`, `mirror`, `resize`, `multmatrix`, `color`, `offset`), all booleans (`union`, `difference`, `intersection`), extrusions (`linear_extrude` with twist / scale / slices, `rotate_extrude`), `hull`, `minkowski`, `projection`, list comprehensions, `for`, `if`, `intersection_for`, `function`, `module`, `render`, string ops, math functions, `include` + `use`, `color()` with per-face RGB propagation.

Partial (🟡) — `$fa` / `$fs` are defined so libraries reading them work, but our primitive builder tessellates from `$fn` (or `fn:` on the primitive), not adaptively.

Not supported (❌) — `import()` of external STL/OBJ/DXF (wasm sandbox has no filesystem — pass bytes via `bin:` instead), `surface()` heightmap import, `$vpr` / `$vpt` / `$vpd` / `$vpf` viewport variables (n/a — no viewport at eval time), font-dependent text glyphs beyond basic Latin-1 skeletons.

See `crates/maquette-scad/FIDELITY.md` for the exhaustive per-feature tracker.

Design notes

Why Manifold? — the previous kernel we used (`csgrs` 0.20.1, BSP-based) left about 20% of faces open on any non-trivial boolean (bores, notches, partial overlaps). Manifold guarantees watertight, correctly-triangulated output. Every mesh this crate emits has zero open edges, verified on the full test corpus.

Why compile in-crate? — a `.scad`-then-shell-out design would need a filesystem, a working OpenSCAD binary on the host, and IPC across a process boundary. The wasm plugin has none of that. Ingesting `.scad` text ourselves keeps the whole compile deterministic under Typst's sandbox — same input, same PDF, cross-platform.

Two wasm modules. — `maquette-scad.wasm` (this crate) compiles geometry to PLY. `maquette.wasm` (the sibling plugin) renders the PLY. They only exchange the PLY blob; you can also send `scadypst(...)`'s output to `maquette-gltf` via a `.gltf` wrapper if you want PBR shading on procedural geometry.